

Track P lepton K-type identification — 3 generations from N_c substrate-mechanism

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Track P lepton K-type identification

1. Question

Standard Model has 3 lepton generations: e, μ, τ + corresponding neutrinos ν_e, ν_μ, ν_τ . WHY 3 generations?

Substrate-mechanism candidate: $N_c = 3$ BST primary structurally forces 3 generations.

2. Lepton K-type framework

Per A_sub v0.9 super-quiver + Half-Integer Axis G v0.2: - Leptons are fermion K-types (half-integer m): $\sigma_{BF} = -1$ - Leptons are color-singlet (don't carry SU(3) color charge): substrate's color SU(3) fiber TRIVIAL on lepton K-types

2.1 Color-singlet projection

Per A_sub Section 4: color SU(3) acts as TRANSVERSE fiber over each K-type node. Leptons are color-singlet $\rightarrow$ color fiber trivializes; lepton K-type lives on Wallach K-type backbone WITHOUT color content.

2.2 3-generation structure

Substrate-mechanism candidate: 3 lepton generations emerge from substrate's 3 color directions being PROJECTED OUT. Each color direction (R, G, B per substrate's $N_c = 3$) contributes a lepton generation when color is removed:

- Generation 1 (electron e): lowest-Casimir Pin(2) fermion K-type ($V_{(1/2, 1/2)}$) projected by R color direction
- Generation 2 (muon μ): $V_{(1/2, 1/2)}$ projected by G color direction
- Generation 3 (tau τ): $V_{(1/2, 1/2)}$ projected by B color direction

All three at SAME Wallach K-type ($V_{(1/2, 1/2)}$) but with different color-projection-direction labels.

Substrate-mechanism for 3 generations: $N_c = 3$ color directions $\rightarrow$ 3 lepton generations via color-singlet projection.

3. Lepton masses via Macdonald specialization

Per multi-phase quiver v0.6: substrate Macdonald at $(q=2, t=\alpha)$.

Lepton masses at different generations could emerge from: - Same Wallach K-type $V_{(1/2, 1/2)}$ (same partition $\lambda = (1/2, 1/2)$) - DIFFERENT Macdonald evaluation per color-projection-direction - Each generation: $P_{(1/2, 1/2)}(x; q=2, t=\alpha)$ evaluated at generation-specific variable specialization

This is FRAMEWORK candidate; multi-week explicit super-Macdonald computation per generation.

3.1 Standard BST mass ratios

Per RATIFIED BST: - $m_\mu/m_e = 207$ per T2003 ($N_c^2(\text{rank}^2 \cdot C_2 - 1)$) at 0.11% - $m_\mu/m_e \approx 206.8$ per T190 ($(24/\pi^2)^6$) at 0.035% - $m_\tau/m_e = 49.71$ per Vol 1 Ch 11 at 0.05%

Substrate-mechanism for generation-specific mass ratios: emergent from substrate's super-Macdonald specialization at different color-direction projections.

4. Neutrino flavor structure

Standard: ν_e, ν_μ, ν_τ neutrino flavors per lepton generation. Substrate-mechanism: same 3-generation structure inherited; neutrinos as Majorana spinors (per Half-Integer Axis G v0.2 + Pin(2) cover).

5. Connection to Five-Absence (NO sterile neutrinos)

Per Five-Absence Principle: NO sterile neutrinos beyond 3 generations.

Substrate-mechanism: 3 color directions = 3 lepton generations. NO 4th color direction means NO 4th lepton generation, NO sterile ν_4 .

Substantive structural reading: Five-Absence “NO sterile neutrinos” is substrate's $N_c = 3$ limit on generation count.

6. Honest scope (Cal #27 STANDING)

What's RATIFIED: - $N_c = 3$ BST primary - A_{sub} gauge fiber structure (color SU(3) transverse) - σ_{BF} fermion sublattice (Pin(2) Z_2 grading) - T2003 + T190 m_μ/m_e mass ratios - Five-Absence NO sterile neutrinos

What's FRAMEWORK candidate: - 3 lepton generations from $N_c = 3$ color directions via color-singlet projection - Generation-specific Macdonald specialization for mass differences - Substrate-mechanism for “NO sterile” via $N_c = 3$ limit

What's INTERPRETIVE / OPEN: - Explicit color-direction-projection mechanism for distinguishing $e/\mu/\tau$ at same Wallach K-type - Generation-specific Macdonald evaluation explicit computation

Cal #29 STANDING audit pass: substrate-derivation framework from RATIFIED $N_c = 3$ + standard color-singlet projection; not back-fitted to 3-generation count.

— Lyra, lepton K-type identification + 3-generation substrate-mechanism filed. Substrate's $N_c = 3 \rightarrow 3$ lepton generations via color-singlet projection on 3 color directions. Same Wallach K-type $V_{\pm}(1/2, 1/2)$ for all generations with different color-direction labels. Five-Absence NO sterile = $N_c = 3$ generation limit.